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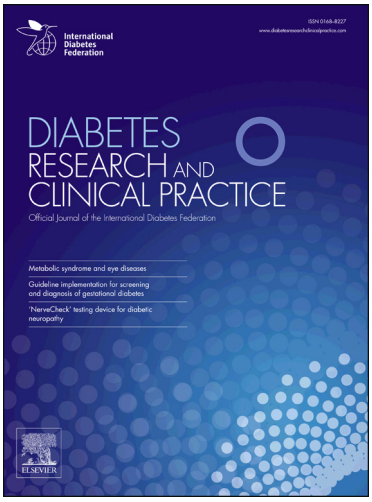

Effects of a lifestyle education program on glycemic control among patients with diabetes at Kigali University Hospital, Rwanda: A randomized controlled trial

Amendezo Etienne, Walker Timothy David, Karamuka Vincent, Robinson Brian, Kavabushi Patrick, Ntirenganya Cyprien, Uwiragiye Joseph, Mukantagwabira Donatille, Bisimwa Jeanne, Uwintwali Marie Henriette, Umulisa Henriette, Niyomwungeri Scholastique, Ndayambaje Bernard, Dusabejambo Vincent, Bavuma Charlotte

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EFFECTS OF A LIFESTYLE EDUCATION PROGRAM ON GLYCEMIC CONTROL AMONG PATIENTS WITH DIABETES AT KIGALI UNIVERSITY HOSPITAL, RWANDA: A RANDOMIZED CONTROLLED TRIAL

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ABSTRACT

Aim

Evidence to show whether lifestyle intervention programs are beneficial for patients with diabetes in resource-limited countries is lacking.

The present study assessed the additional efficacy of a structured lifestyle education program, as compared to the current standard of diabetic care in Rwanda.

Methods

251 consecutive adult patients attending a tertiary diabetic care practice were randomly assigned to either an intervention group (standard of care plus monthly lifestyle group education sessions of 45 minutes duration) or to a control group.

The primary outcome was between-groups difference in glycated hemoglobin (HbA1c) observed after 12-months follow up.

Outcome measures in the intervention and control groups were compared using the ANCOVA test with a two-sided significance of 5%.

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Results

Of the 251 subjects recruited, 223 were included in the analysis; of whom 115 were assigned to the intervention group, and 108 to the control group.

After 12-months, the median HbA1c levels reduced by 1.70 (95% CI: -2.09 to -1.31); p<0.001) in the intervention group; and by 0.52 (95% CI: -0.95 to -0.10); p= 0.01) in the control group.

The difference in HbA1c reduction between the intervention and control groups was statistically significant (p<0.001) after adjustment for subjects' age, sex, education level, BMI, diabetes duration and diabetic medications.

Conclusions

This study demonstrated that a structured lifestyle group education program for people with diabetes is an attractive option in a resource-limited setting, as it showed significant benefits in improved glycemic control over a 12-month period.

Key words: Diabetes Mellitus; Structured Lifestyle Education Program; Rwanda

Trial registration

ClinicalTrials.gov: NCT02032108

1. BACKGROUND

In Sub-Saharan Africa (SSA), as elsewhere in the world, the increasing prevalence of diabetes mellitus and other non-communicable diseases adds undue strain to health systems already burdened by high rates of infectious diseases and malnutrition-related health problems (1)(2)(3)(4)(5).

Diabetes in SSA kills people in their most economically productive years, which has long standing negative repercussions on families and wider society (6)(3)(4)(7).

Unfortunately in Rwanda, as in many other developing countries, access to comprehensive diabetic care is still limited (1)(3)(8)(9)(10). Moreover, people living with diabetes in Rwanda and SSA are younger on average and develop disease

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complications earlier than persons with diabetes from the United States and other developed countries (3)(11)(12)(13).

Previous research carried out in developed countries demonstrated that lifestyle modification can have a significant effect on both the development of diabetes and on progression of established disease (14)(15)(16)(17). Furthermore, such interventions have been described to improve both glycemic control and mitigate complications (1).

Direct evidence showing that group-based lifestyle interventions are beneficial for people living with diabetes in resource-limited countries is limited, with the only published trial from South Africa showing only improved blood pressure control(18)(19).

Although access to a healthy diet is often claimed to be the major barrier to implementation of lifestyle based initiatives for diabetes in Africa (20), we hypothesized that under-addressed knowledge gaps among people living with diabetes are more important barriers to improved care than access to healthier diet and other lifestyle measures, and that the level of individual counseling on lifestyle modifications that patients with diabetes receive during clinical encounters is suboptimal, even in a specialized clinic in a tertiary center. We therefore carried out this randomized clinical trial to assess the efficacy of an additional lifestyle education program delivered through group diabetic counseling, compared to the current tertiary standard of diabetic care in Rwanda.

2. METHODS

This study was an un-blinded, parallel-group, randomized controlled trial evaluating the impact of a structured lifestyle modification program on diabetes control, compared with tertiary standard of care treatment, in Rwanda. The study was conducted at the outpatient facility of the largest tertiary center in Kigali, Rwanda.

Study participants were adults, age 21 years or older, who had been diagnosed with diabetes mellitus at least three months prior to enrollment into the study. Pregnant

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patients or those with severe co-morbid illnesses that would impede their functional capacity during follow up were excluded.

People living with diabetes and followed up at this hospital were invited to take part of the study. The study team recruited those who voluntarily accepted to enter the study. The recruited participants were randomly assigned to an intervention group (local standard of care plus monthly lifestyle group counseling & education sessions lasting between 45 and 60 minutes) or to a control group (standard care for diabetes in our setting, which consists of monthly medical follow up and individual counseling on dietary habits and lifestyle change, delivered by attending physicians and/or nutritionists as required).

Each study participant was followed up for 12-months, starting from the first encounter, which was the day of recruitment; and ending at the last encounter, which was the day of blood collection for 12-month HbA1c testing. For those randomized to the intervention group, the first session was organized within three weeks from their first encounter/recruitment into the study.

Of note, the study team was not allowed to change patients' medications at any point during the trial; this was left at the attending doctor's discretion.

For the intervention group, the study team administered monthly group counseling sessions. The study team was composed of five physicians, four nurses, three nutritionists and two psychologists who had received training for the purpose of this study.

Group education sessions focused on: setting balanced diabetic diet (including salt reduction, fruit and vegetable consumption) based on available local food and patients' financial capability; regular physical activity; cessation of smoking and alcohol abuse; adherence to medications and to regular medical follow up; diabetic complications screening and treatment; self-management of hypoglycemia and hyperglycemia and stress management.

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Participants assigned to the interventional group also received educational pamphlets translated in the local language (Kinyarwanda) to complement the oral information received during monthly sessions. The latter were developed by the study team, adapted into the local culture and context from international guidelines on diabetes and from prior research protocols evaluating similar interventions (1)(9)(21)(22)(23)(24).

To standardize teaching materials and methods, facilitators attended three training sessions prior to study initiation. Additionally, the principal investigators held monthly meetings with the team of educators and data collectors to provide feedback on the study progress and ensure quality control in the administration of the intervention.

Data were collected using a pre-established Case Report Form. Demographic data as well as clinical and laboratory data were collected at enrollment, and at six and 12-month follow up visits. HbA1c testing was done at enrollment and every three monthly.

The primary outcome of this study was between-groups difference in HbA1c. Secondary outcomes were differences from baseline in HbA1c after one-year follow-up, between-groups and baseline-to-12 months differences in systolic and diastolic blood pressure, body mass index (BMI), and fasting blood glucose.

Hospital technicians, who were blinded to the participants' group assignments, collected this data. It was not possible to blind patients to their treatment group assignment due to the nature of the intervention. Of note, there was no change in the outcomes from the initial design to the end of the study.

2.1. Sample size and statistical analysis

The study was designed to have a statistical power of 80% power to detect a reduction in HbA1c ≥ 0.40 in the intervention group compared to the control group during a median follow up of one year with a two-sided significance level of $p < 0.05$. To achieve this, we estimated that we needed 108 subjects per arm, thus a total of 216 study participants. We purposively recruited a total of 251 subjects to account for an

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anticipated number of subjects who would be lost-to-follow up. The HbA1c reduction of 0.40 is based on prior research of similar interventions (25)(26)(27).

The analysis was performed on an intention-to-treat basis. All study participants with a HbA1c test result available at 12 months follow-up were analyzed, according to their initial treatment assignment. Mean (or median if population not normally distributed) values of demographic data in the intervention and control groups were compared using the Student's t-test (for parametric data) or Wilcoxon test (for non-parametric data) with a two-sided significance of 5%. A one way ANCOVA analysis was conducted to determine statistical difference between control and intervention groups on outcome measures of interest, namely HbA1c, systolic and diastolic blood pressure, fasting blood sugar and weight; after controlling for age, sex, profession, education level, duration of diabetes, the household size, comorbid conditions, presence of any diabetic complication, marital status, physical activity level, fasting blood sugar, type of diabetic medication, weight, and number of sessions attended in a multivariate analysis.

Statistical analysis was performed using STATA software version 11 (StataCorp. Texas, USA).

2.2. Ethics

The study was approved by the Research and Ethics Committee at Kigali University Teaching Hospital. Each participant included in the study gave a written informed consent before enrollment.

Participants were not paid for their participation in the study. Group education sessions were scheduled on routine clinic days in order to minimize extra visits to the hospital. HbA1c and urine albumin/creatinine ratio laboratory tests were provided to all participants at no cost.

3. RESULTS

The trial was stopped once 12-month follow-up on all participants was completed, as planned. Of the 251 participants included in the analysis, data was available for 223 participants; 21 were lost to follow up or were unavailable on scheduled visits for HbA1c testing, four became pregnant and three died (Figure One: Trial Flow Chart).

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Of study subjects who were randomized to the interventional group, 72.1% attended a minimum of eight group educational sessions, while 22.6% attended between three and seven sessions. Of note, two participants who were initially randomized to the control group attended two educational sessions despite not being invited. These subjects were included in the analysis of the control group, as per the intention-to-treat principle. (Figure One).

3.1. Demographic Data

The mean age of study participants was 51.5 (+/-11) years; the majority (71%) were female; 66% had secondary school education or higher; the mean duration of diabetes was 7.3 (+/-6.6) years; 53.4% had a family history of diabetes and 47.2% of women had a personal history of fetal macrosomia. Of all, 21.5% of subjects were recently diagnosed with diabetes (diagnosed between three months and one year prior to enrollment). Forty nine per cent of participants had hypertension; while 28.2% were obese (BMI>/= 30kg/m^2).

The average household size of participants was 6.3 persons per house. The baseline demographic characteristics were not significantly different between the two study groups (Table One).

3.2. Baseline clinical and laboratory characteristics

The median baseline HbA1c for all study participants was 8.98%. The median urine albumin/creatinine ratio was 27.4 µg/mg with 16.9% of subjects having abnormally elevated levels (Table Two). The median blood pressure was 136.6/80.7 mmHg and median BMI was 27.9 Kg/m^2. The baseline clinical and laboratory characteristics did not significantly differ between the two groups. (Table Two).

3.3. Primary and Secondary Outcomes

The median HbA1c significantly decreased in both the intervention and the control groups (from 9.19% at baseline to 7.49% at 12-months visit; and from 8.74% at baseline

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to 8.21% at 12-months visit; respectively). Moreover, the between group difference in change in HbA1c at 12-months was statistically significant (p <0.001) (Table Two).

This statistical difference was maintained after controlling for age, sex, profession, education level, duration of diabetes, the household size, comorbid condition, presence of any diabetic complication, marital status, physical activity level, fasting blood sugar, type of diabetic medication, weight or BMI, and number of sessions attended in a multivariate analysis.

Furthermore, at 12-month follow up visit, between group differences in fasting blood glucose, systolic blood pressure, diastolic blood pressure and weight were all statistically different (p <0.001, 0.005, 0.02, <0.001 respectively), owing to greater improvement of those variables in the intervention group, compared to the control group (Table Three).

There was an increase in the number of study participants who met the target HbA1c (</=7%) at the 12-month visit in both the control and intervention groups (HbA1c goal was achieved in 15.7% of all study participants at baseline and in 38.6% at 12-months). A greater proportion of subjects in the intervention group achieved their goal HbA1c at 12-months compared to the control group (48.8% vs 28.7%, p=0.003) (Figure Two).

4. DISCUSSION

This randomized controlled trial evaluated the impact of a monthly, contextualized, group lifestyle education program on the glycemic control of tertiary-care patients with diabetes in Rwanda, demonstrating significantly greater improvements in glycemic control and other important clinical parameters at 12 months in the intervention group as opposed to the control group.

Baseline data show that prior to study initiation the degree of glycemic control in study participants was poor (only 15.7% of all study participants had a baseline HbA1c

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</=7%). Similar figures have been reported in Kenya (8)(10), suggesting that our study enrolled a population typical of that seen at tertiary centers in East Africa. The figures indicate that diabetic care in Africa is still far behind that in developed countries such as USA, where at least 55% of patients with diabetes are reported to have a HbA1C of </= 7%(28) (29).

This observed high proportion of patients with diabetes in Rwanda with poor glycemic control, correlates with the previously reported high prevalence of diabetic complications in Rwanda (12). These figures are alarming and suggest that diabetes will have an increasingly devastating human and economic toll, if the trends are not reversed in the near future. This in turn points to the need for novel therapeutic interventions, to improve care and reduce complications.

Our study demonstrates that improvements in diabetic care in Rwanda, and other challenging resource-limited, clinical settings are possible, without a huge increase in per-patient spending. Group lifestyle modification education programs, similar to programs already used in developed countries, can have dramatic impact on diabetes control in Africa, and require very little financial outlay. Overall, at a one-year follow up, the number of subjects achieving HbA1c goal of </= 7% had increased from 15.7% to 38.7%. There were significantly more patients achieving HbA1c goals in the intervention group, compared to the control group; indicating a significant contribution of the group counseling sessions to improved diabetes management in Rwanda.

In particular, the striking decrease in HbA1c by 1.7% in the intervention group compared with 0.52% in the control group is highly clinically significant for diabetic patients in Rwanda. Prior studies in other populations have demonstrated that each 1% reduction in HbA1c over 10 years is associated with a risk reduction of 21% in diabetes related deaths, 14% in myocardial infarctions, and 37% in micro-vascular complications(32)(33). It is likely that this benefit will carry over in African populations, although definitive research in our population has not yet been performed.

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Other trials comparing group and individual behavior modification programs for diabetes control have reported on significantly better outcomes with group counseling which we did also observe in this study. Xavier et al. in the Look AHEAD Trial observed significant reduction in HbA1c and cardiovascular risk factors (blood pressure, cholesterol, triglycerides, weight) among participants in a group lifestyle modification program compared to individual counseling over one year follow up, although no difference in cardiovascular outcomes was observed (34). Trento et al. similarly reported that hyperglycemia; hypertension and dyslipidemia were all better controlled with group educational programs than individualized or no counseling (35). In our study, people with diabetes randomized to the interventional arm had better profiles on hyperglycemia, hypertension and weight after 12-months follow up.

The novel training, which diabetic care providers involved in the study received prior to study initiation, likely increased provider knowledge on the role of diet and exercise in diabetic management in both arms of the study and could partly explain the improvements in HbA1c among both participants from interventional and control groups.

In addition, this being the first intervention of its kind in Rwanda and the novelty of conducting such a study in Rwanda may have primed the participants to take an unprecedented interest in their own disease management and this as well can explain the observed outcome improvements in both the intervention and control groups in the present study. Of note, although about one fifth of subjects were recently diagnosed with diabetes (within one year), the duration of diabetes did not influence the statistical analysis results in a multivariate model.

Lifestyle modification programs in SSA have been previously evaluated for their role in diabetes prevention (9). To our knowledge, however, the current study is one of the few that looked at the role of lifestyle modification programs on glycemic control in established diabetes in SSA. The observed effects of diabetic education on glycemic control; the high rates of participant retention in the intervention group (72.1% of participants in the interventional arm attended a minimum of eight group educational

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sessions, which is similar to previously reported data (16)(25)(27)), as well the tremendous enthusiasm towards lifestyle modification approach to diabetic management expressed by both patients and providers illuminates the great potential of such programs in routine diabetic care in Rwanda.

Importantly, the team of group session educators was mostly composed of primary care health professionals, thereby signaling the feasibility of such an intervention program and its applicability in other similar settings and even in lower level health facilities. Previous studies have shown that lifestyle intervention programs conducted by primary care teams have beneficial effects on glycemic control, other CV risk factors as well as overall well-being of people living with diabetes (16)(26)(27).This may be particularly relevant for African countries where the majority of NCD care must be provided at primary level healthcare facilities. While further studies need to be done, it is conceivable that such patients are less educated about their diabetes, and thus might accrue even greater benefits than those seen in our study population.

4.1. Study strengths and limitations

The principal strengths of the present study are its randomized, prospective design and the fact that it addresses lifestyle group education for diabetes care in Rwanda - a country where the number of health professionals is limited and where such interventions had not been previously evaluated while they can save time spent on individual education. The present study provides initial clues that behavior modification programs may be useful in the public health management of non-communicable disease in SSA.

The study has some limitations. Although the intention to treat principle was followed, and loss-to-follow-up was limited, there was a small differential in the loss to follow-up between intervention and control arms (8 vs 20 patients respectively). It is however unlikely that this differential loss would have dramatically changed the study conclusions, given the robust and consistent nature of the findings.

Secondly, the patients and providers were not blinded to the intervention or to the results of the primary and secondary outcomes and this might have potentially

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introduced bias into the results, although the use of objective clinical markers for study endpoints should help to limit this bias, and the lack of blinding and quarantining of the intervention group from the control group is likely to favor the null hypothesis.

While it is previously described that type 2 diabetes patients are those likely to show improvements with lifestyle intervention programs, our study included people with both diabetes type one and type two. We are aware that this could have introduced heterogeneity into the study population, and that our results may not be generalizable to all people with type 1 and type 2, but we did not have means to accurately differentiate the two groups, given the known complexities of diabetic subtypes in Africa.

In conclusion, results of the present study demonstrate that a structured lifestyle education program is both feasible and effective in a resource limited setting such as Rwanda.

The statistically significant trends of glycemic profile improvements in the interventional group over the control group, as well as the observed optimistic outcomes suggest that similar interventions may be effectively implemented in other SSA countries with similar and even lower level health facilities.

4.2 Conflicts of interests

All authors declare no conflicts of interest.

4.3. Funding sources

This study received funding from Sanofi Aventis and Kigali University Teaching Hospital's Department of Research. Sanofi provided funds for HbA1c and urine albumin/creatinine testing while KUTH paid for development of study materials. Neither institution has been involved in the analysis of study data, the interpretation of the study results or the decision to submit the article for publication.

4.4. *Acknowledgements*

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4.5. Authors' contribution:

EA: study design, data entry and analysis, interpretation of results and manuscript writing

BC: study design, data collection and manuscript writing

TW: study design, data analysis, interpretation of results and manuscript writing

BR: Study design and data collection

KV: Data collection, data entry, and manuscript writing

NC, KP, MD, UJ, MC, UMH, UH, BJ, NS, NB, and DV: Subjects recruitment and conducting education sessions.

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Figure One. Trial flow chart

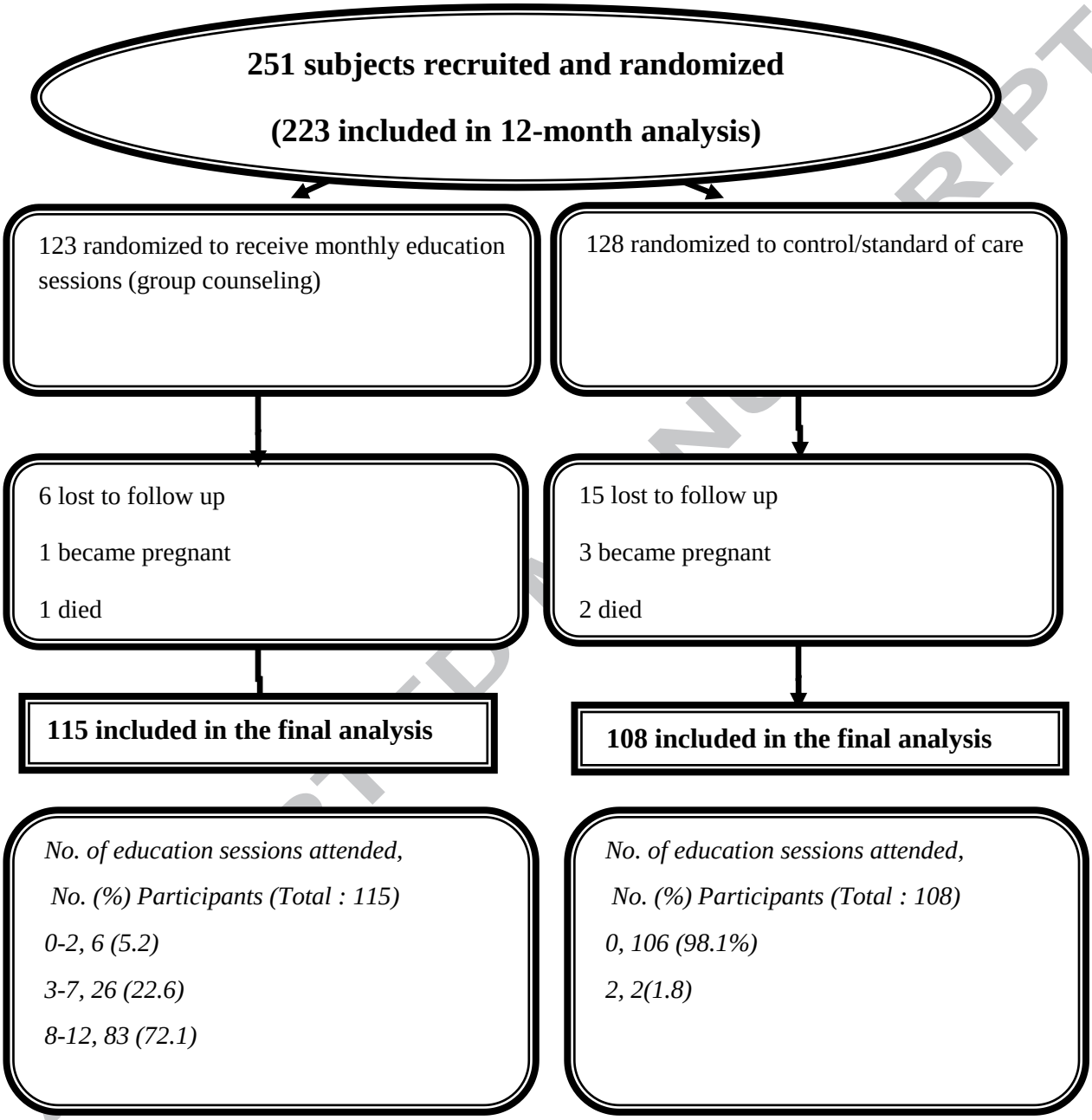

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Table One. Study participant demographic characteristics at baseline

	All (n=251)	Intervention (n=123)	Control (n=128)	P-value
Mean Age (years)	50.9 (+/- 10.9)	51.4 (+/- 10.9)	50.5 (+/- 11.0)	0.60
Sex (% females)	69.3	67.5	71.1	0.53
Education (%)				0.15
None	4.0	1.6	6.3	
Primary school	31.9	29.3	34.4	
Secondary & technical school	41.4	46.3	36.7	
University	22.7	22.8	22.7	
Duration of diabetes (years)	7.0 (+/-6.4)	6.8 (+/- 5.7)	7.2 (+/- 7.0)	0.33
0 – 5 yrs	51.4	49.6	53.1	
6 – 10 yrs	22.3	26.0	18.8	
11 – 15 yrs	16.3	17.1	15.6	
More than 15 yrs	10.0	7.3	12.5	
Household size	6.3 (+/-2.6)	6.0 (+/-2.5)	6.7 (+/-2.5)	0.4
Family history of diabetes (% positive)	53.4	52.8	53.9	0.67
Personal history of fetal macrosomia (% positive)*	47.2	53.1	41.8	0.3
Comorbid conditions (%)				0.83
None	35.1	33.3	36.7	
Arterial hypertension	**49.0**	**48.0**	**50.0**	
HIV infection	5.6	7.3	3.9	
Asthma&other respiratory disease	3.2	3.3	3.1	
Chronic gastritis/Peptic Ulcer Disease	2.4	3.3	1.6	
Viral hepatitis B or C	1.2	0.8	1.6	
Other	3.6	4.1	3.6	
History of tobacco smoking (%)	28.3	30.1	26.5	0.7
History of alcohol consumption (%)	66.1	63.4	68.7	0.14

**Percentages were computed out of 165 female participants who responded to the question*

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Table Two. Baseline clinical and laboratory characteristics of all study participants

	All (n=251)	Intervention (n=123)	Control (n=128)	P-value
HbA1c (%)	8.98 (8.6-9.3)	9.13 (8.7-9.7)	8.87 (8.3-9.3)	0.52
% Subjects acheiving HbA1c goal (</=7%)	15.7	5.8	9.9	0.06
Systolic BP (mmHg)	136.6 (133.1-139.1)	135.3 (130.9-138.5)	137.8 (132.8-142.0)	0.65
Diastolic BP (mmHg)	80.7 (78.3-81.3)	80.0 (77.5-81.7)	81.3 (77.9-82.2)	0.12
Weight (kg)	75.6 (73.8-77.8)	75.2 (73.1-78.8)	75.9 (72.8-78.6)	0.38
BMI (kg/m^2) #	27.9 (27.0-28.5)	27.6 (26.7-28.7)	28.2 (26.8-29.0)	0.7
<18.5*	2.0	2.4	1.6	
18.5-24.9*	25.1	26.8	23.4	
25-29.99*	44.6	45.5	43.8	
>/=30*	**28.3**	**25.2**	**31.3**	
Waist circumference (cm)	96.2 (94.4-98.2)	95.9 (94.2-99.1)	96.3 (93.2-98.9)	0.73
Total Cholesterol (mmol/l)	4.3 (4.04-4.30)	4.4 (4.0-4.6)	4.2 (3.8-4.4)	0.11
Triglycerides (mmol/l)	1.8 (1.7-2.0)	1.8 (1.6-2.1)	1.8 (1.6-2.0)	0.4
Serum Creatinine (µmol/l)	79.7 (76.8-84.2)	79.8 (74.7-82.5)	79.6 (73.5-86.5)	0.58
Urine albumin/creatinine ratio	27.4 (16.7-37.8)	24.8 (12.7-34.2)	30.0 (12.7-49.4)	0.51
No microalbuminuria (<30 µg/mg) ****	83.1	85.1	82.6	
Microalbuminuria (>30 µg/mg)****	16.9	14.9	17.4	
Diabetic medications*				0.77
Insulin only	24.5	22.1	26.7	
Metformin only	13.3	15.5	12.6	
Sulfonylurea only	10.0	8.8	10.6	
Insulin+metformin	8.8	11.0	11.6	
Insulin+Sulfonylurea	2.0	3.3	2.1	
Metform+Sulfonylurea	40.6	43.4	37.7	
Other	0.8	2.2	1.1	

**Given values in percentage*

 # BMI was calculated as weight in kilograms divided by height in meters squared.

***Percent out of 207 subjects on whom the test was performed*

BMI: Body Mass Index. BP: Blood Pressure

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Table Three. Changes in clinical and laboratory outcomes at 12 months (n=223)

	Intervention (n=115)				**Control (n=108)**				**Mean Diff. p-value ****
	Baseline median (95%CI)	12 month med (95%CI)	Diff.	p-value*	Baseline median (95%CI)	12 month median (95%CI)	Diff.	p-value*	
HbA1c	**9.19 (8.7-9.6)**	**7.49 (7.22-7.76)**	**-1.7**	**<0.001**	**8. 74 (8.32-9.15)**	**8.21 (7.88-8.53)**	**0.52**	**0.015**	**<0.001**
Fasting Glucose (mmol/L)	8.4 (7.77-9.04)	6.9 (6.45-7.36)	-1.49	<0.001	8.19 (7.62-8.76)	9.02 (8.18-9.87)	+0.81	0.13	<0.001
Systolic BP (mmHg)	135.1 (131.7-138.6)	131.8 (129-134.6)	-3.31	0.058	137.3 (133.3-141.3)	140.8 (137.1-144.4)	+3.5	0.04	0.005
Diastolic BP (mmHg)	80.1 (78.2-81.9)	77.0 (75.3-78.8)	-3.02	0.003	81.1 (79.0-83.3)	81.6 (79.6-83.6)	+0.5	0.65	0.02
Weight (kg)	75.4 (72.9-77.8)	73.7 (71.3-76.1)	-1.7	<0.001	76.4 (73.8-79.0)	77.3 (74.6-79.9)	+0.88	0.055	<0.001

Diff = difference in values between baseline and 12 months.

** p-value for within group comparison (baseline vs 12-month follow up for each of the two study groups)*

***Mean diff. p-value = p-value for between randomization (intervention vs control) group difference*

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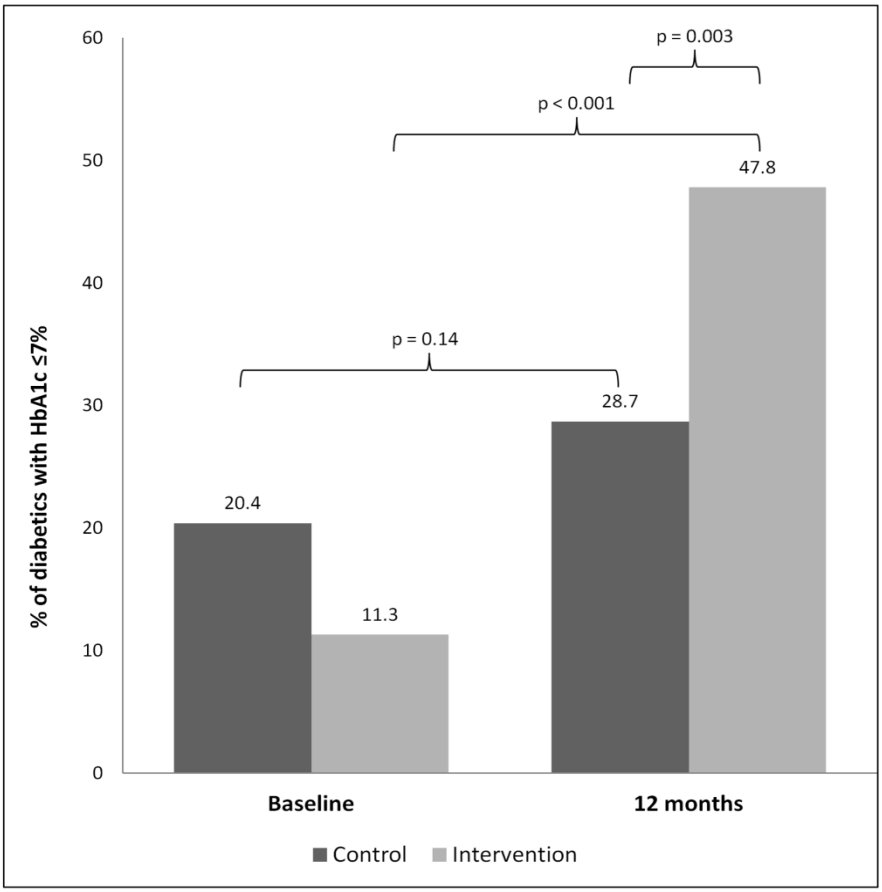
Figure Two. Percentage of study subjects achieving HbA1c ≤ 7 at baseline and at 12-month follow up



NB: “p”: p-values by chi-square test, for both inter-group and over-time comparisons.

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HIGHLIGHTS

- Benefits of a structured lifestyle education program for people with diabetes living in resource-limited settings were measured in this single-centre randomized control trial
- The structured lifestyle group education program was both feasible and effective, with high levels of attendance
- More subjects randomized to the lifestyle group education program achieved their HbA1c goals at 12-months, compared to controls receiving usual care

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